

Examen Practico aplicaciones descentralizadas parcial 2 (UNIDAD 2 y 3)

Números de control: 19100199, 19100163 Fecha: 16/04/2024

Calificación: _____ Nombres: Juan Hernandez Rosas,
Gerardo Alberto Cruz Rojo

Conteste las siguientes preguntas:

1.-¿Cuáles son las características de una cuenta externa?

- Llave publica del usuario
- Solo pueden operar conociendo la llave privada.
- Si se compramos algo en Ether o si iniciamos una accion con un smart contract, se deberia almacenar en este tipo de cuenta.

2.-¿Cuáles son los campos de una transacción ?

- Direccion del solicitante.
- Direccion del remitente
- Cantidad de ether a transferir en Wei.
- Datos de entrada
- Cantidad inicial de gas. Si este sobrepasa en EVM, mostrara una excepcion y hara rollback.
- Firma digital, que confirma que el solicitante es el que hace la transaccion.
- Precio por unidad de gas

3.-Describa el proceso del ciclo de vida del ether

- Minería de ether:

Es generado a traves del proceso de mineria, durante el cual un grupo de mineros compiten en un grupo para crear y agregar nuevos bloques. Si se tiene exito, un minero obtiene su recompensa, el cual es un numero de Ether coins. Si los bloques se agregan cada 15 seg su demanda aumenta.

- Transferencia de ether:

Una vez generado el ether, se almacena en la cuenta externa del minero. Esta transferencia se puede hacer por los mineros los cuales lo pueden hacer mediante una cartera

- Intercambio de ether:

El ether tiene un valor, esto no se transfiere gratuitamente entre cuentas. Se transfiere a cambio de bienes y servicios intercambiados. Se puede comprar ether de dueños individuales.

- Almacenado de ether:

Se almacena en una cuenta. Se pueden manejar las cuentas utilizando varios métodos, cada uno con su propia conveniencia y seguridad. Los mas comunes y convincentes es a través de

carteras utilizando aplicaciones de escritorio las cuales permiten transferir Ether sencillamente. La mas segura es a través de generar una llave privada física o en una billetera física que significa generar la llave privada de manera offline y guardarla en un pedazo de papel.

Realice lo siguiente utilizando la Blockchain y realice un reporte con cada una capturas necesarias para comprobar los puntos

1. Buscar la siguiente información

1. Información del nodo

```
C:\WINDOWS\system32\cmd.exe  Windows PowerShell
modules: admin:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.0 personal:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d or type exit
> admin.nodeInfo
{
  "enode": "enode://c298b03b3454b0798ecd3b8912eb47dc5e6fda772bbcd078c1be1fe9175f61df09f61e59cb079eeadef9f88c3b40ae0f34eabf64212e34868a00bd5d6f31d9e127.0.0.1:30303",
  "enr": "enr:-aaQ2_Kv2bq6cX4e03yid7Lzd0B0:F4N4wRuhwih_vZc2GaHgZACFDFA2vK00dt3Bc1XoZXBwtCMLXF07WbtE6teuGAY434FqHg2V0aMr3hPHz2ue0GnrLgaLkgnY0gaLwhH3AAAGJc2Vjc011NasxcoQPCmLA7Wf57Apjs07iRLrR9xeb9p3K7z0eHG-H-4XX2HYRzbfwIN0Y3CCd1-DdWRegnZf",
  "id": "baf6fc45691f9a1ca68af07cf03c86ae448188f07152fa7222afbfa0c5bbfdd",
  "ip": "127.0.0.1",
  "listenAddr": "[*]:30303",
  "name": "eth/v1.15.14-stable-2bd6bd01/windows-amd64/go1.21.6",
  "ports": {
    "discovery": 30303,
    "listener": 30303
  },
  "protocols": {
    "eth": {
      "config": {
        "berlinBlock": 0,
        "byzantiumBlock": 0,
        "cancunTime": 1706655072,
        "chainId": 11155111,
        "constantinopleBlock": 0,
        "daoForkSupport": true,
        "eip150Block": 0,
        "eip155Block": 0,
        "eip158Block": 0,
        "ethash": {},
        "homesteadBlock": 0,
        "istanbulBlock": 0,
        "londonBlock": 0,
        "mergeNetsplitBlock": 1735371,
        "muirGlacierBlock": 0,
        "petersburgBlock": 0,
        "shanghaiTime": 1677557088,
        "terminalTotalDifficulty": 17000000000000000,
        "terminalTotalDifficultyPassed": true
      },
      "difficulty": 131072,
      "genesis": "0x25a5cc106eea7138acab33231d7160d69cb777ee0c2c553fcddf5138993e6dd9",
      "head": "0x25a5cc106eea7138acab33231d7160d69cb777ee0c2c553fcddf5138993e6dd9",
      "network": 11155111
    },
    "snap": {}
  }
}
```

2. Cantidad de peernodes

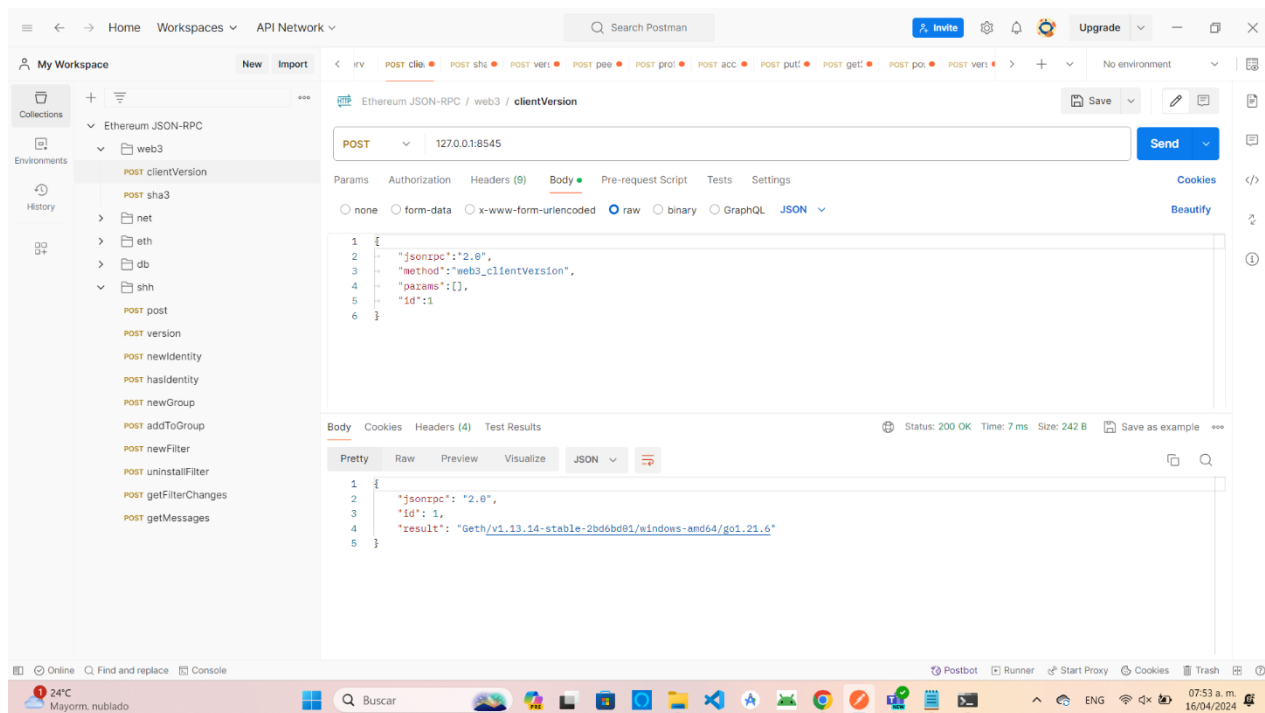
```
C:\WINDOWS\system32\cmd.exe  Windows PowerShell

config: {
  berlinBlock: 0,
  byzantiumBlock: 0,
  cancunTime: 1706655072,
  chainId: 11155111,
  constantinopleBlock: 0,
  daoForkSupport: true,
  eip150Block: 0,
  eip155Block: 0,
  eip158Block: 0,
  ethash: {},
  homesteadBlock: 0,
  istanbulBlock: 0,
  londonBlock: 0,
  mergeNetsplitBlock: 1735371,
  muirGlacierBlock: 0,
  petersburgBlock: 0,
  shanghaiTime: 1677557088,
  terminalTotalDifficulty: 17000000000000000,
  terminalTotalDifficultyPassed: true
},
difficulty: 131072,
genesis: "0x25a5cc106eea7138acab33231d7160d69cb777ee0c2c553fcddf5138993e6dd9",
head: "0x25a5cc106eea7138acab33231d7160d69cb777ee0c2c553fcddf5138993e6dd9",
network: 11155111
},
snap: {}
}
>
> net.peerCount
1
```

3. Conversiones de ether utilizando sus números de control

```
> web3.toWei(19100199, 'ether')
"19100199000000000000000000000000"
> web3.toWei(19100163, 'ether')
"19100163000000000000000000000000"
>
```

4. 3 comandos de preferencia utilizando el JSON-RPC API Modulo Web3 – ClientVersion



Modulo Web3 – sha3

The screenshot shows the Postman interface with a workspace named "Ethereum JSON-RPC". The selected collection is "web3", and the specific endpoint is "sha3". The request is a POST to "127.0.0.1:8545". The body is a JSON object:

```
{  "jsonrpc": "2.0",  "method": "web3_sha3",  "params": ["0x68656c66f20776f726c64"],  "id": 64}
```

. The response status is 200 OK, with a time of 4 ms and a size of 250 B. The response body is:

```
{  "jsonrpc": "2.0",  "id": 64,  "result": "0x47173285a8d7341e5e972fc677286384f082f0ef42a5ec5f93bbfa254cb01fad"}
```

Modulo net – versión

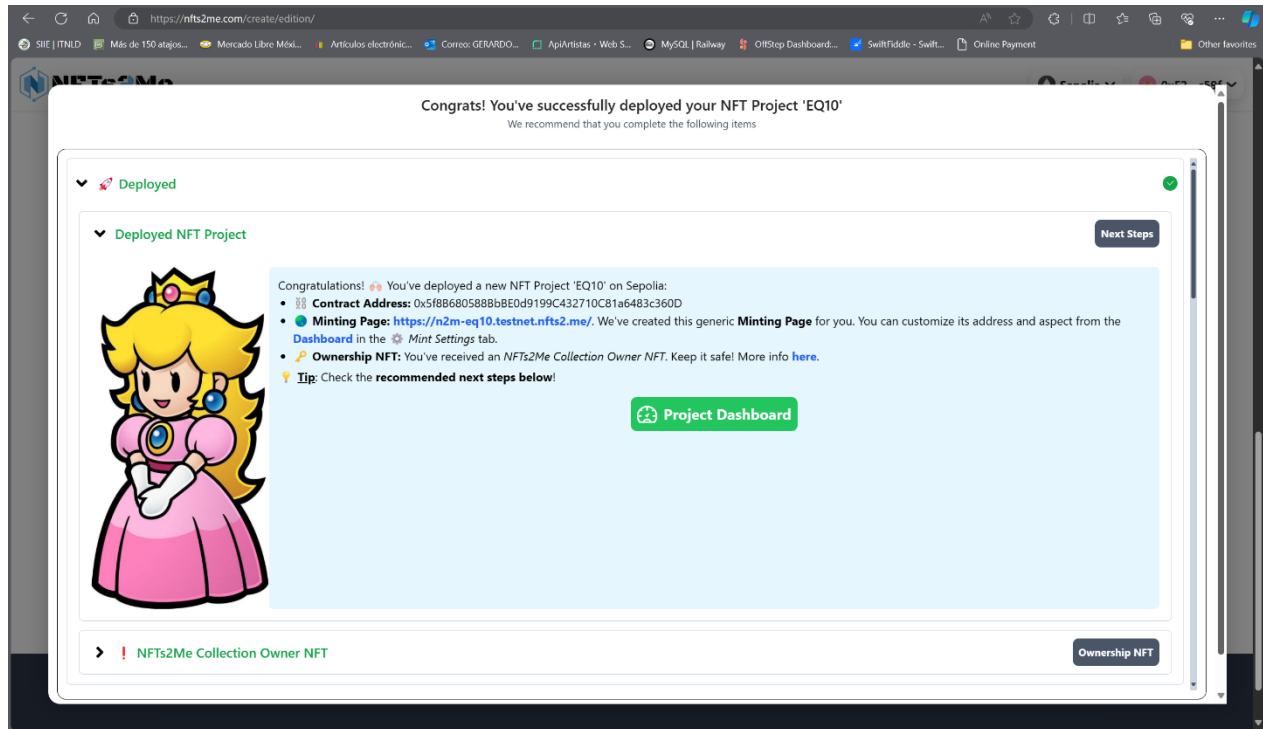
The screenshot shows the Postman interface with a workspace named "Ethereum JSON-RPC". The selected collection is "net", and the specific endpoint is "version". The request is a POST to "127.0.0.1:8545". The body is a JSON object:

```
{  "jsonrpc": "2.0",  "method": "net_version",  "params": [],  "id": 67}
```

. The response status is 200 OK, with a time of 3 ms and a size of 202 B. The response body is:

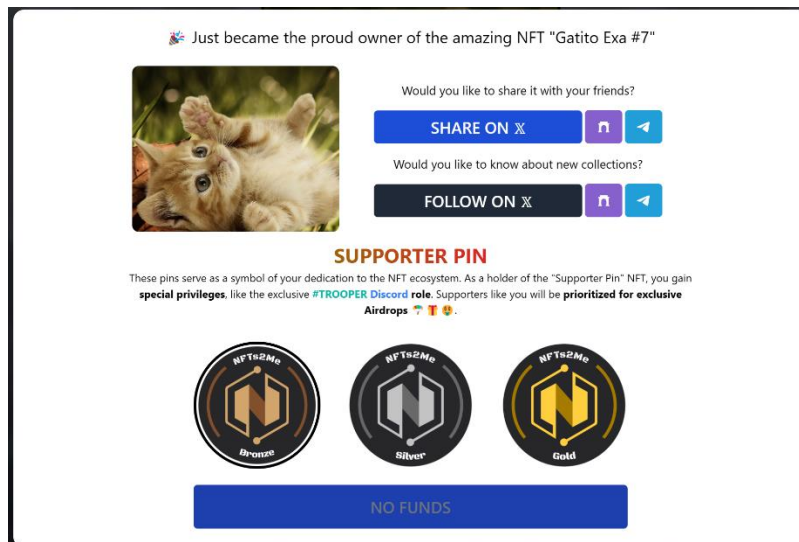
```
{  "jsonrpc": "2.0",  "id": 67,  "result": "11159111"}
```

2. Utilizando Metamask realizar lo siguiente:

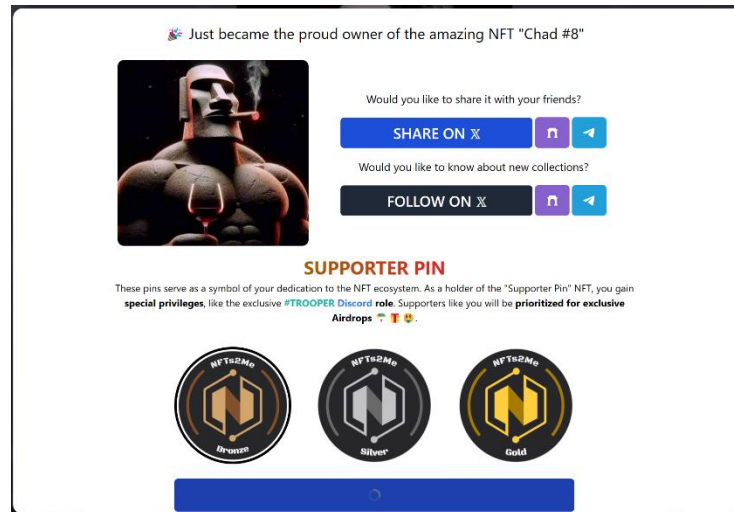


1. Minar un NFT De cada equipo

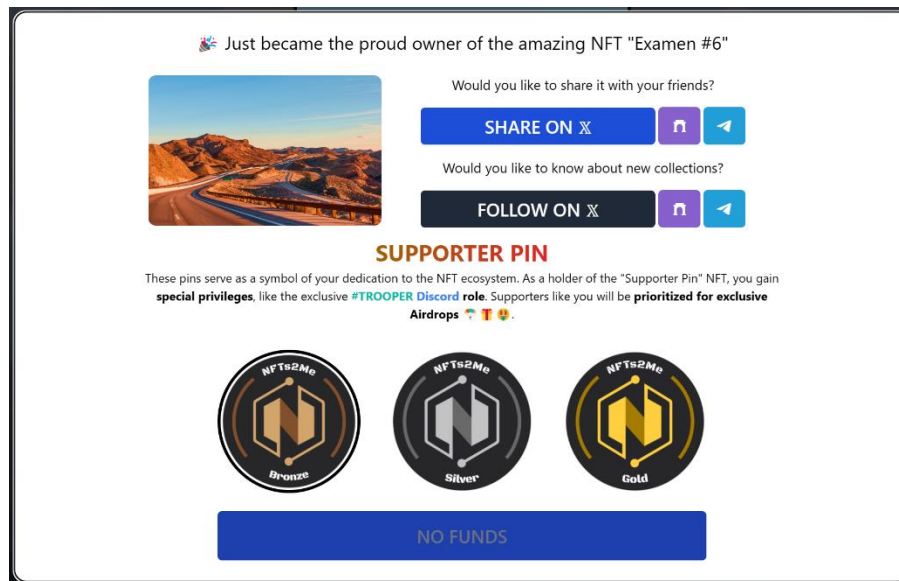
EQ1:



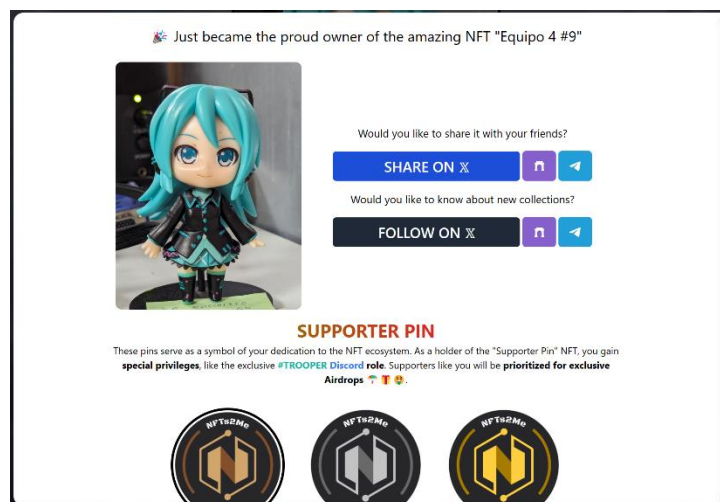
EQ2:



EQ3:



EQ4:



EQ5:

🎉 Just became the proud owner of the amazing NFT "Equipo5 #10"



Would you like to share it with your friends?

SHARE ON  

Would you like to know about new collections?

FOLLOW ON  

SUPPORTER PIN

These pins serve as a symbol of your dedication to the NFT ecosystem. As a holder of the "Supporter Pin" NFT, you gain **special privileges**, like the exclusive **#TROOPER Discord role**. Supporters like you will be **prioritized for exclusive Airdrops** 🎁 🎁 🎁.

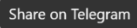


NO FUNDS

EQ6:

🎉 Just became the proud owner of the amazing NFT "Equipo6_nft1 #7"



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SHARE ON  

Would you like to know about new collections?

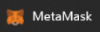
FOLLOW ON  

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 Confirm Transaction on Ethereum

EQ7:

🎉 Just became the proud owner of the amazing NFT "Memardo #7"



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SHARE ON X  

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FOLLOW ON X  

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NO FUNDS

EQ8:

🎉 Just became the proud owner of the amazing NFT "ExamenDescentralizadas2024 - Equipo 8 #10"



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SHARE ON X  

Would you like to know about new collections?

FOLLOW ON X  

SUPPORTER PIN

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NO FUNDS

EQ9:

🎨 Just became the proud owner of the amazing NFT "Mr.Fresh #9"



Would you like to share it with your friends?

SHARE ON  

Would you like to know about new collections?

FOLLOW ON  

SUPPORTER PIN

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NO FUNDS

Eq11:

🎨 Just became the proud owner of the amazing NFT "Examen segunda unidad #10"



Would you like to share it with your friends?

SHARE ON  

Would you like to know about new collections?

FOLLOW ON  

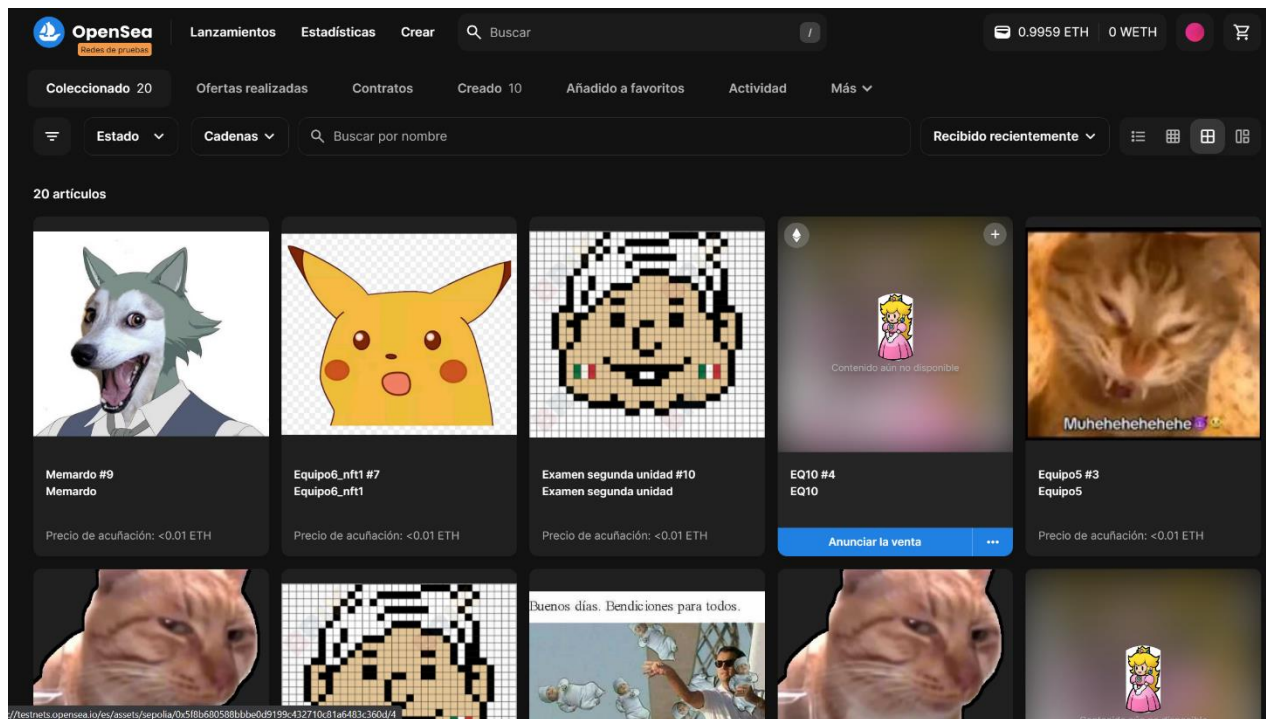
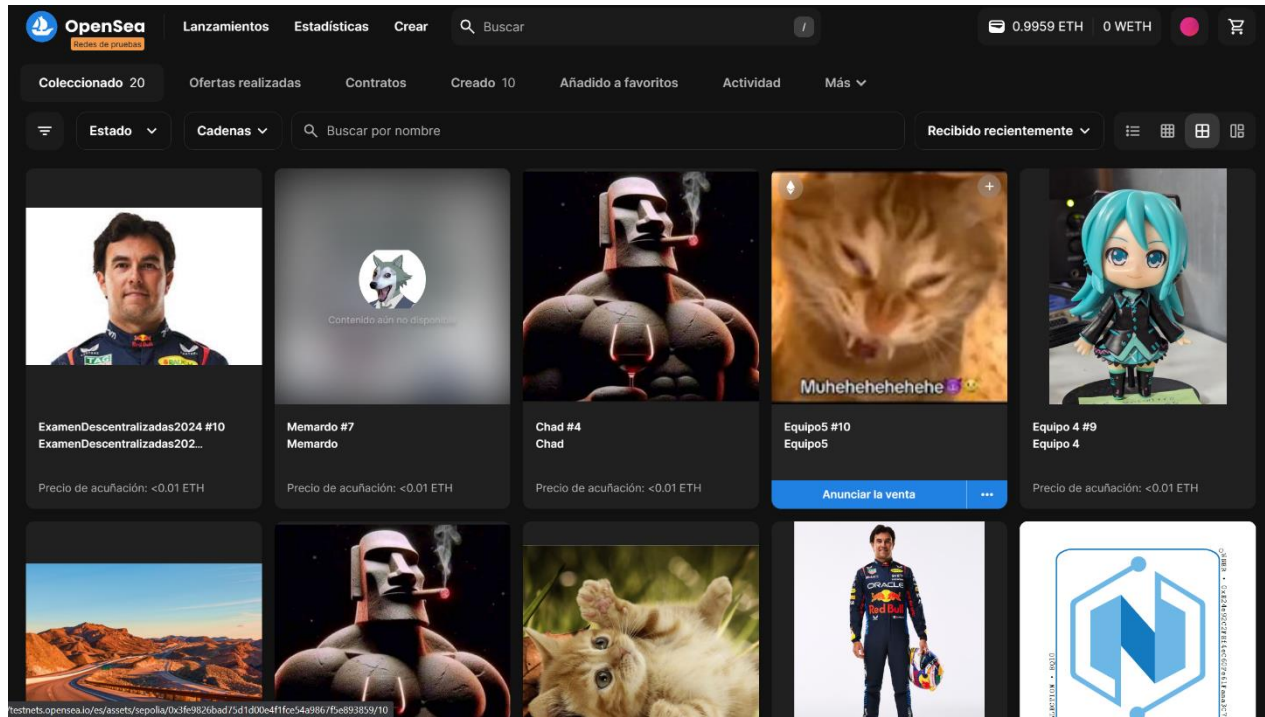
SUPPORTER PIN

These pins serve as a symbol of your dedication to the NFT ecosystem. As a holder of the "Supporter Pin" NFT, you gain **special privileges**, like the exclusive **#TROOPER Discord role**. Supporters like you will be **prioritized for exclusive Airdrops** 🎁 🎁 🎁.



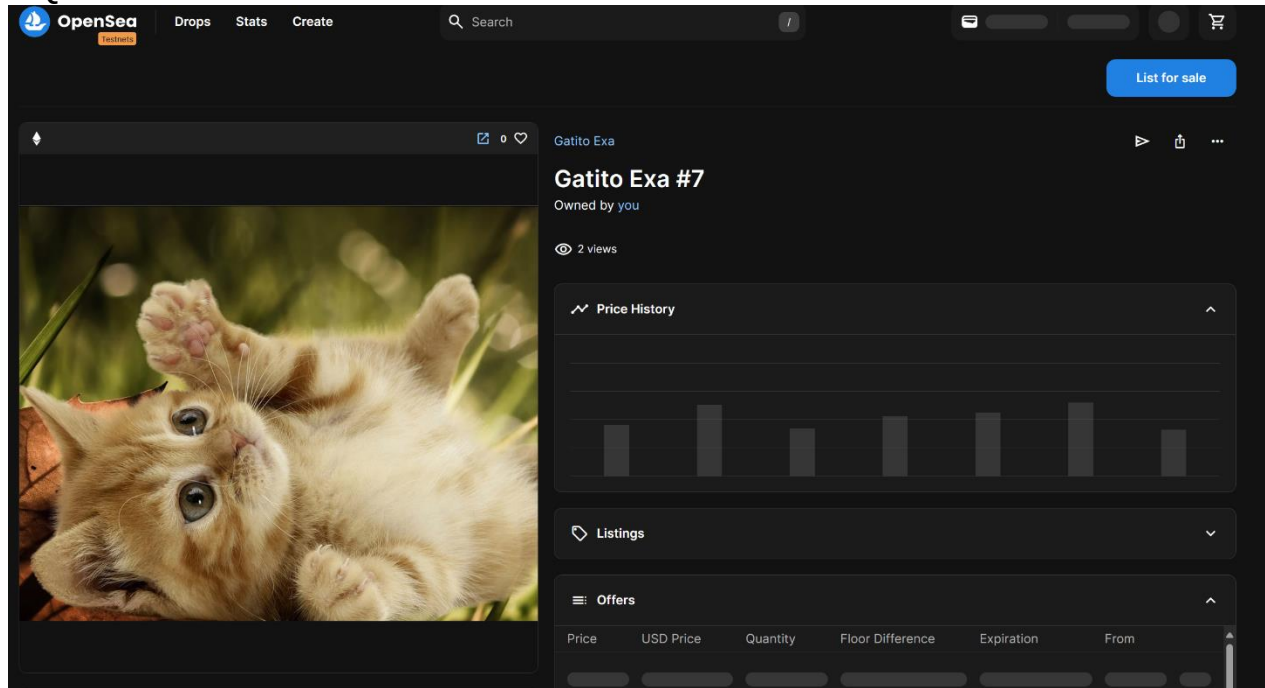
MetaMask
Confirmed Transaction

2. Recibir 1 NFT de cada equipo

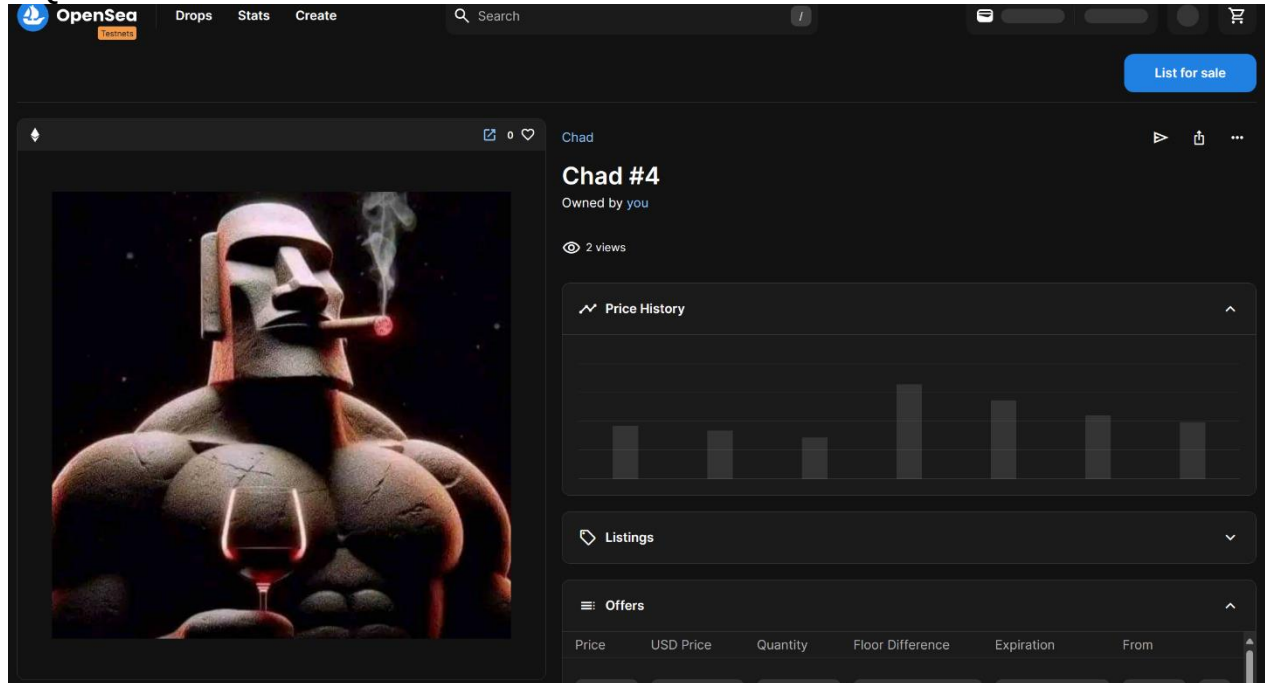


3. Mostrar sus NFT en OpenSea

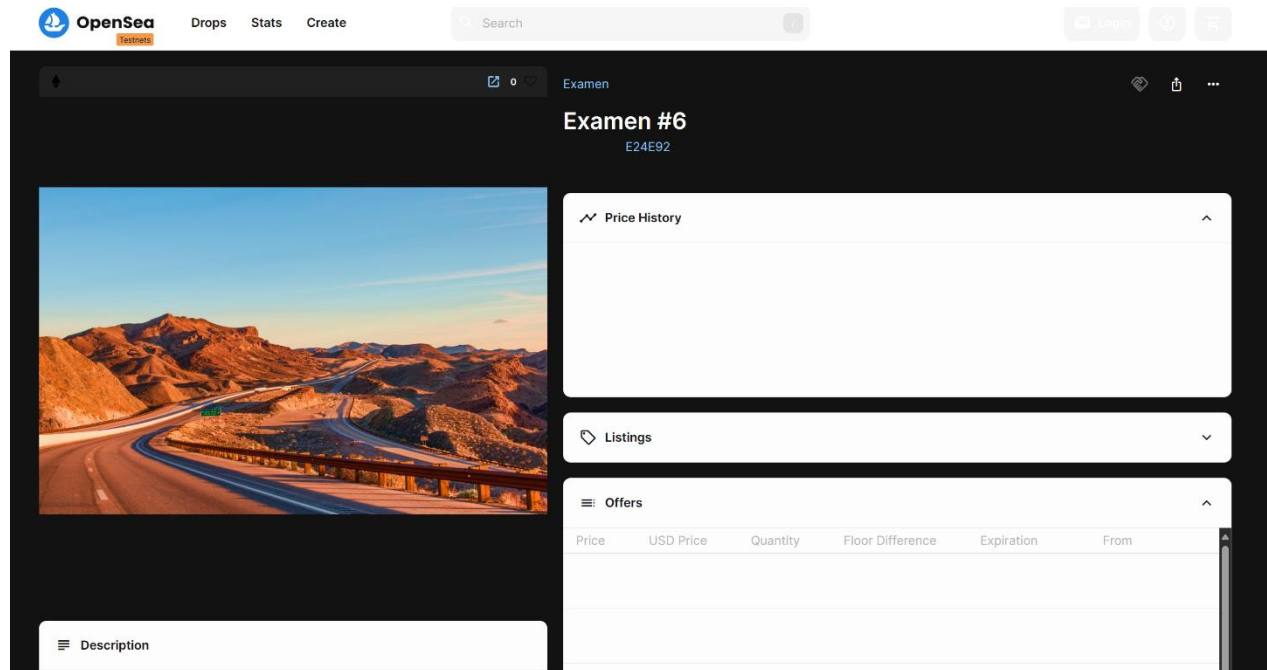
EQ1:



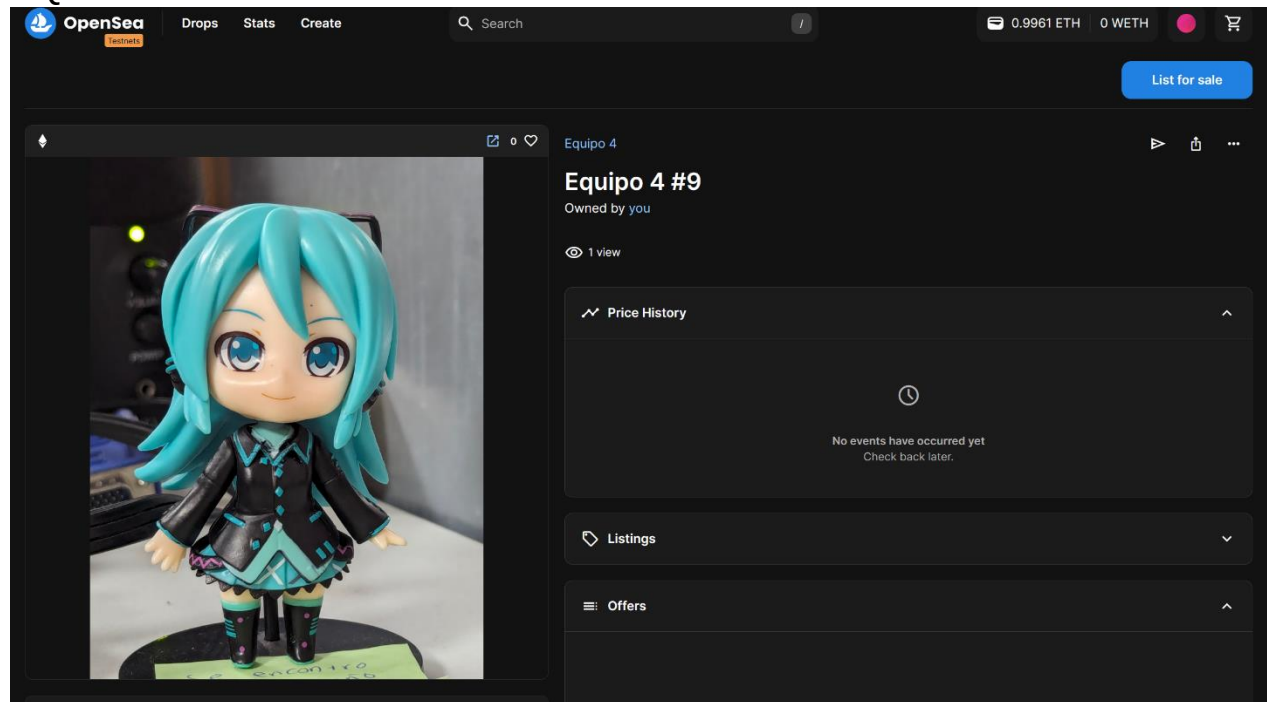
EQ2:



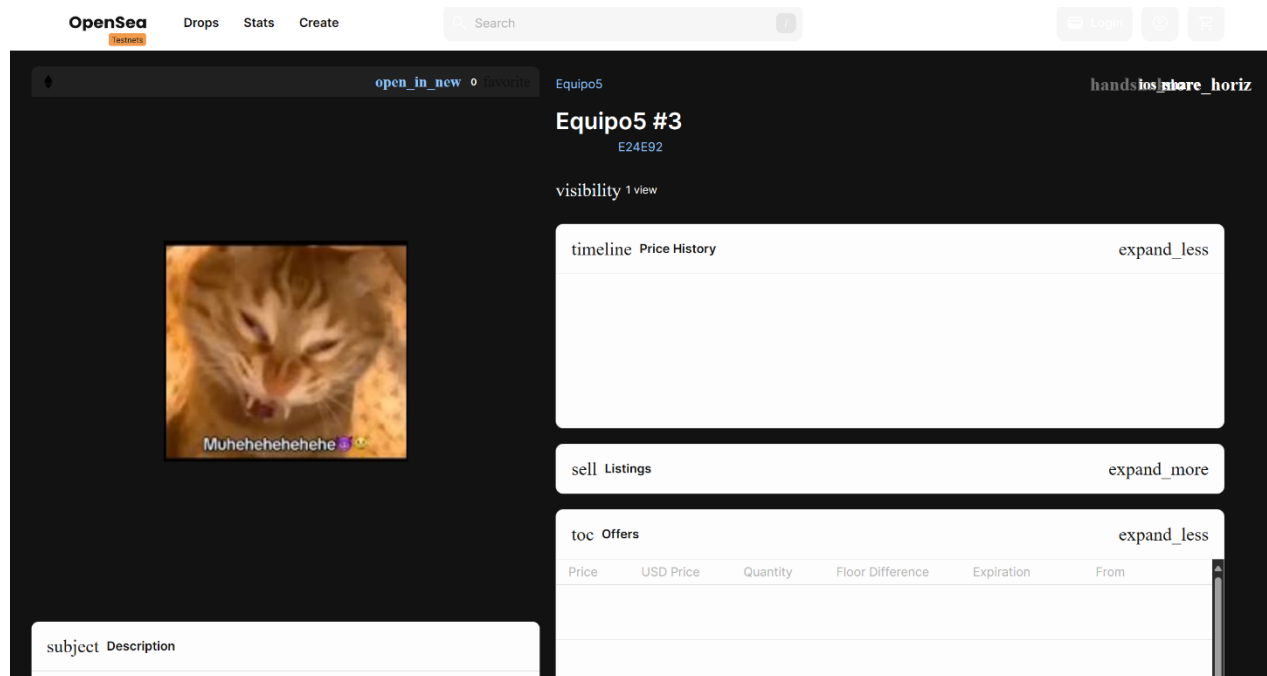
EQ3:



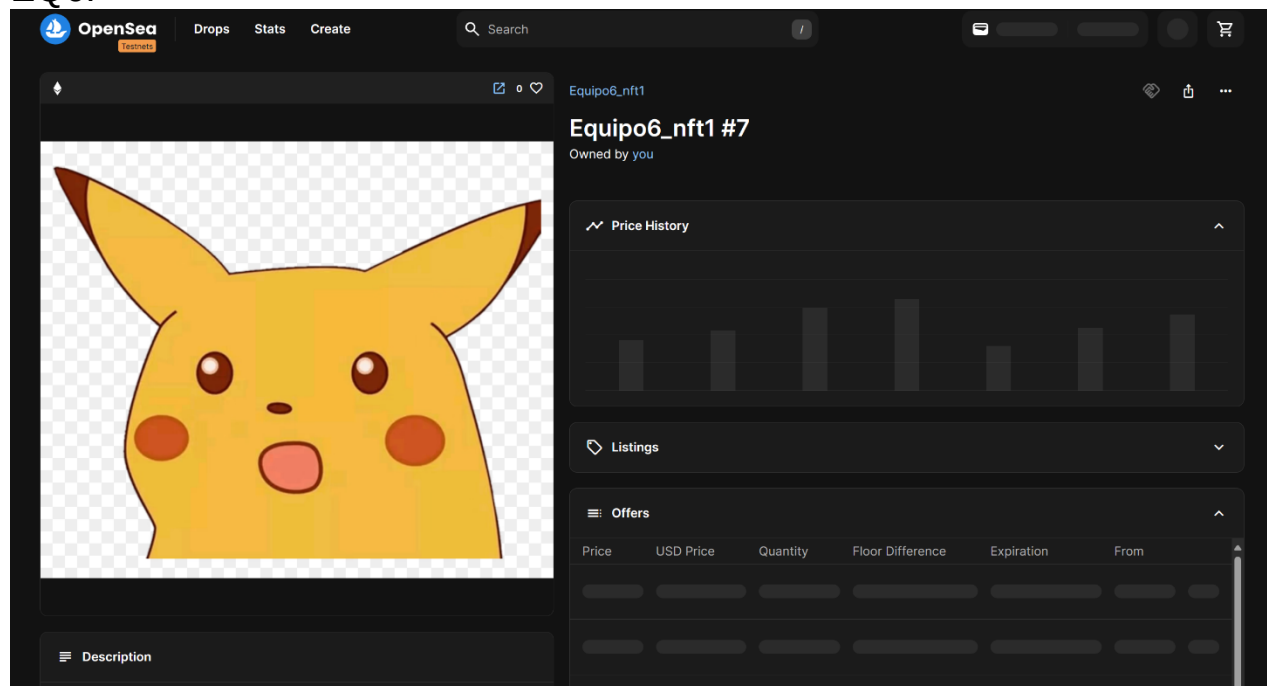
EQ4:



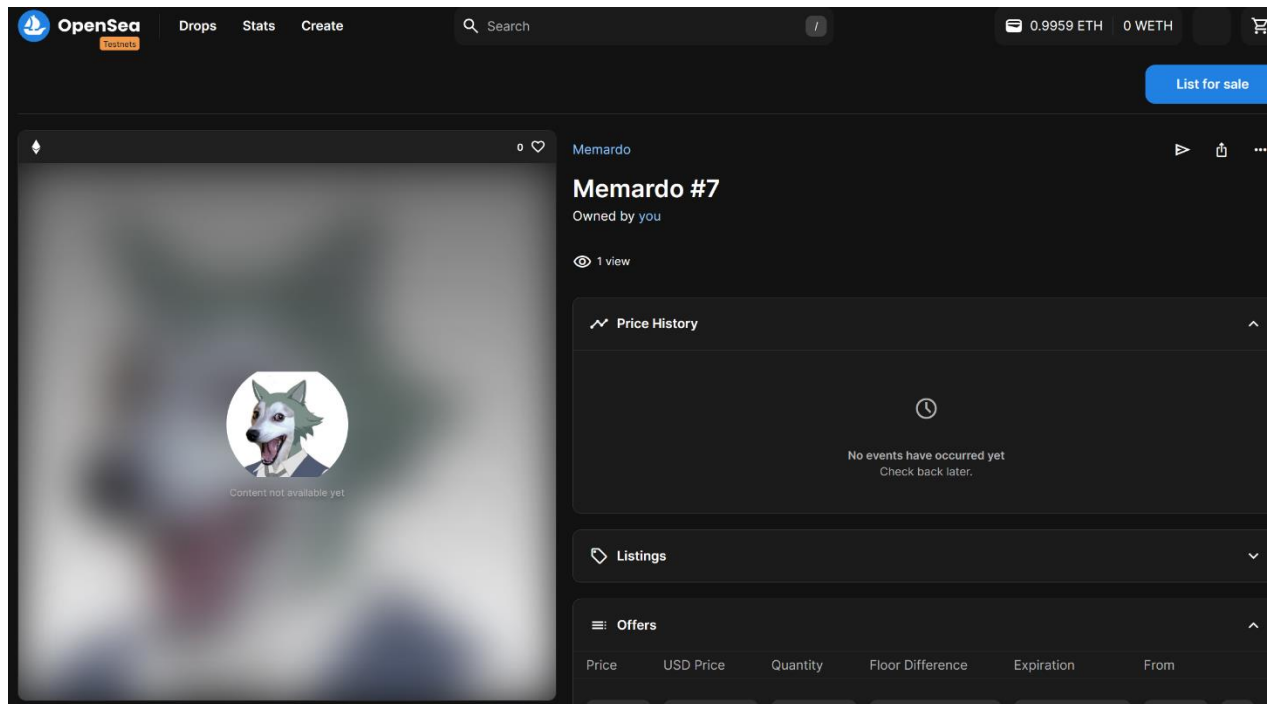
EQ5:



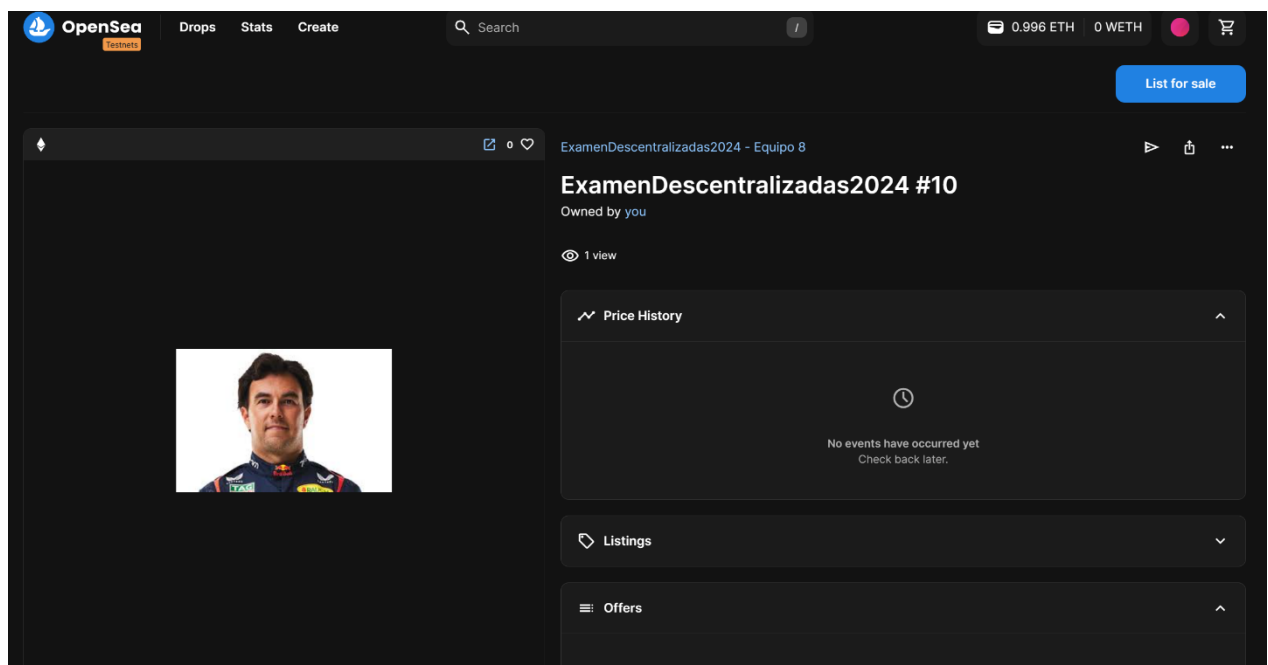
EQ6:



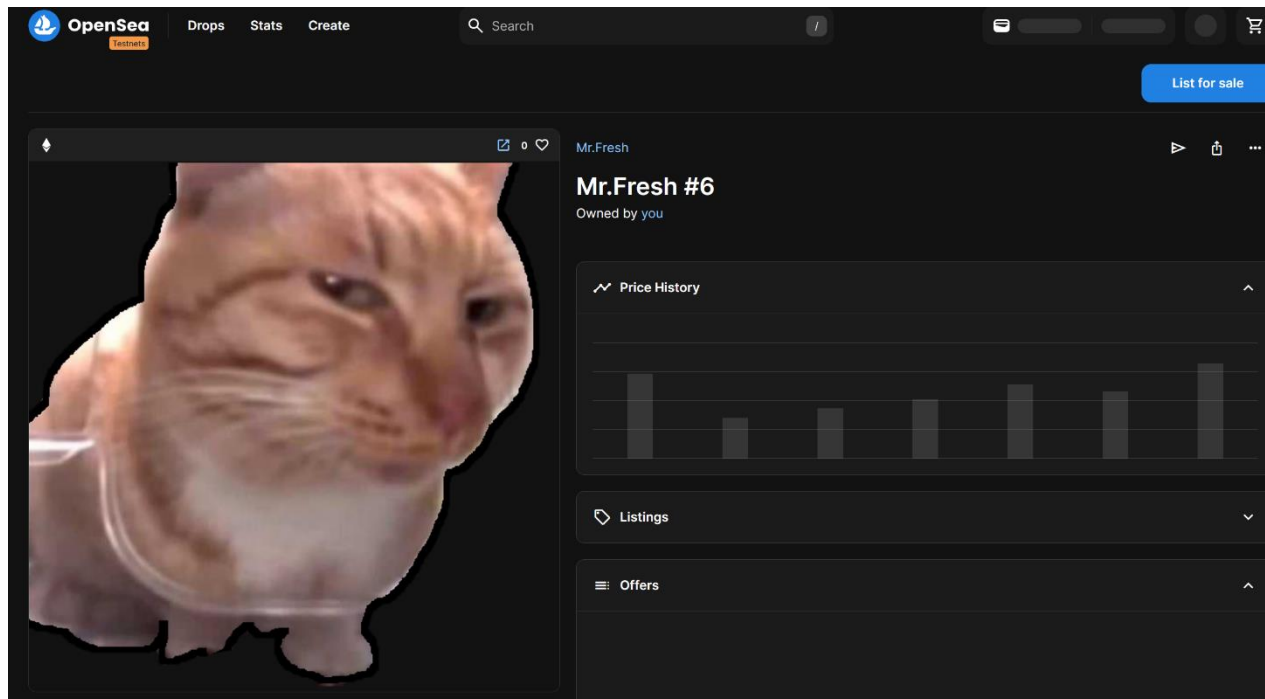
EQ7:



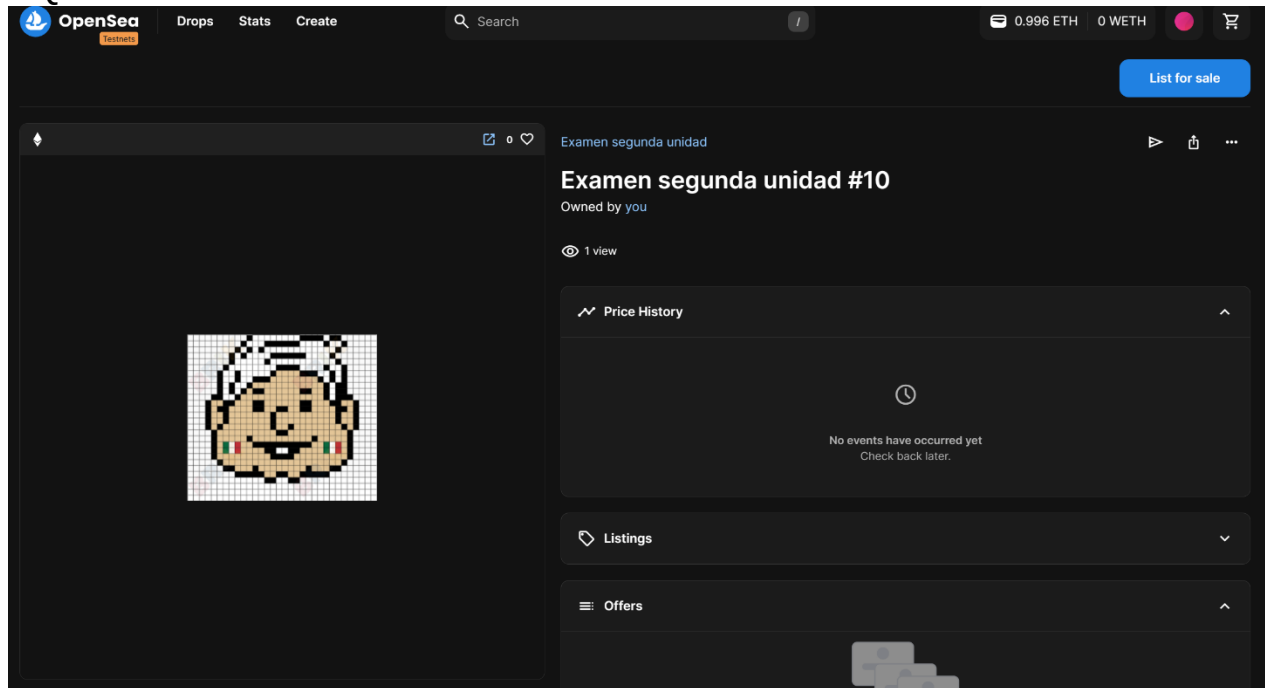
EQ8:



EQ9:



EQ11:



4. Hacer un deploy de un contrato SimpleCoin y repartir 100 + su numero de equipo coins a la cuenta de cada equipo
5. Mostrar el balance de coins de al menos 5 cuentas
6. Mostrar las transacciones utilizando el explorador de la blockchain